PRODUCT REQUIREMENTS DOCUMENT & EXECUTION PLAN

PIXEL-Srishti: Agentic Geospatial AI Assistant

Strategic Convergence of Two SIH 2026 Problem Statements:
1. **SIH26167 (ISRO/SAC):** SatQuery AI (Agentic Vision-Language Assistant)
2. **SIH26015 (MoRD/DoLR):** Geospatial Visualization for Watershed Development

1. Executive Summary & Convergence Strategy

PIXEL-Srishti is a strategic convergence of ISRO's SatQuery AI and the Ministry of Rural Development's Watershed monitoring initiative. Rather than treating these as separate builds, PIXEL-Srishti uses one agentic, vision-language backbone as the engine and points it at two paying customers.

The Strategic Scoring Advantage:

- Judges evaluating **PS26167** will see a live, non-benchmark use case beyond academic datasets.
- Judges anchored on **PS26015's** watershed outcomes will see a technically rigorous, ISRO-grade AI core rather than a simple, manual GIS dashboard.

2. Mandatory Coverage Checklist (PS26167)

To ensure we hit every judging criteria for ISRO, the orchestrator must execute:

☑ **Single-image VQA** (Mandatory baseline - covered by RS-VQA)

☑ **One additional single-image task** (Grounding/Localizing objects)

☑ **Bi-temporal change description** (Change-VQA via Siamese-network)

☑ **Optical-SAR cross-modal analysis** (Fusion pipeline)

☑ **Agentic orchestration with auditable execution trace** (JSON logs)

3. Core Architecture

The system operates on a 4-tier architecture designed for scalability and intelligence:

1. **Data Ingestion Layer:** Normalizes GeoTIFFs, SRTM, Sentinel, and SRISHTI-DRISHTI tiles using `GDAL/rasterio`. Handles CRS reprojection and channel normalization.
2. **Specialist Model Registry:**
   - *RS-VQA/Captioning:* For scene understanding (LLaVA/BLIP-2 style, LoRA fine-tuned on BigEarthNet).
   - *Text-Guided Grounding:* Localizes natural language queries.
   - *Bi-Temporal Change:* Siamese-network (ChangeFormer).
   - *Thematic Segmentation:* U-Net/DeepLabV3+ for land-use mapping.
3. **Agentic Orchestrator (The Brain):** An instruction-tuned LLM using a LangGraph/Tool-calling pattern. It parses user intent, validates image modality, sequences models, and returns an auditable execution trace. **Graceful Degradation:** If SAR or T2 imagery is missing, the controller halts and explains what's missing rather than hallucinating.
4. **Presentation & Audit Layer:** A React.js frontend featuring an interactive geospatial map (Leaflet) to render thematic layers, plus a machine-readable JSON execution trace that government evaluators can trust.

4. Phased Implementation Plan (Hackathon Sprint)

Based on the optimized build sequencing principle, we will not train every model from scratch. We will prioritize end-to-end orchestrator flow over raw model accuracy for the MVP.

Phase 1: Foundation & Data Layer

- **Backend:** Set up a FastAPI server with endpoints for image upload (GeoTIFF) and chat querying.
- **Geospatial Processing:** Implement utility scripts using `rasterio` and `geopandas` to normalize inputs.
- **Frontend Scaffold:** Initialize React.js split-screen layout (Chat + Map).

Phase 2: Agentic Orchestrator (The Brain) & Auditability

- **Agent Logic:** Implement a LangChain/LangGraph agent.
- **Strict Routing:** Program the agent to check modality compatibility before execution.
- **Audit Trace:** Ensure the agent emits a JSON trace (tool selection, latency, coordinates) for every query.

Phase 3: Specialist ML Pipelines (Execution)

- **Integrate Pre-trained Tools:** Connect lightweight VQA and segmentation tools.
- **Simulated Heavy Pipelines:** For complex Optical-SAR fusion or Bi-temporal change, wrap simulated/mock outputs in the exact same API format while background fine-tuning happens.

Phase 4: LoRA Fine-Tuning (The Core VLM)

- **Targeted Fine-Tuning:** Use parameter-efficient fine-tuning (LoRA) on an open-source remote-sensing VLM (e.g., GeoChat) using BigEarthNet. This is the single highest-impact ML task.

Phase 5: Polish & Deployment

- **Frontend Map Rendering:** Connect the agent's output so GeoJSON/masks overlay instantly on Leaflet.
- **UI/UX:** Apply professional SIH/ISRO-themed CSS styling. Make the Audit Trace visible to judges.